

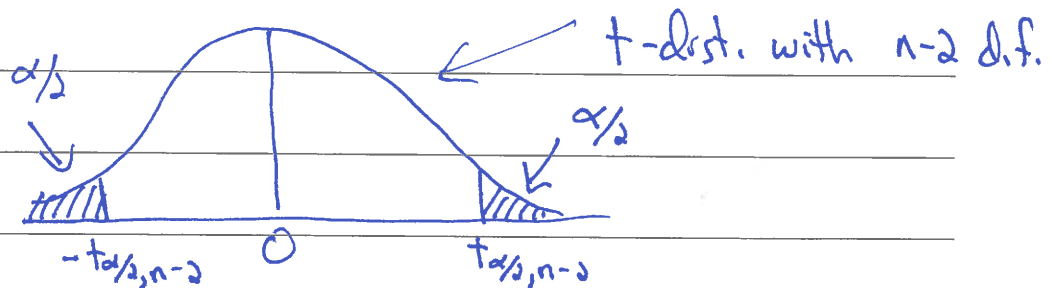
Exercise: Construct a $100(1 - \alpha)\%$ confidence interval for β_1 .

$$\alpha = 0.05 \Rightarrow 95\% \text{ CI}$$

"pivotal approach to constructing a CI"

$$P(-t_{\alpha/2, n-2} \leq T \leq t_{\alpha/2, n-2}) = 1 - \alpha$$

where $t_{\alpha/2, n-2}$ is the $1 - \alpha/2$ percentile of the t -dist. with $n-2$ d.f.



$$P(-t_{\alpha/2, n-2} \leq \frac{\hat{\beta}_1 - \beta_{10}}{SE(\hat{\beta}_1)} \leq t_{\alpha/2, n-2}) = 1 - \alpha$$

$$= P\left(\underbrace{\hat{\beta}_1 - t_{\alpha/2, n-2} SE(\hat{\beta}_1)}_L \leq \beta_{10} \leq \underbrace{\hat{\beta}_1 + t_{\alpha/2, n-2} SE(\hat{\beta}_1)}_U\right) = 1 - \alpha$$

$$P(L \leq \beta_{10} \leq U) = 1 - \alpha$$

(L, U) is a $100(1 - \alpha)\%$ CI for β_{10}

Comments:

① B_{10} is not random in this construction

L, U are random variables

Hence it makes sense to write $P(L \leq B_{10} \leq U) = 1 - \alpha$

We get data. Plug in to calculate $\hat{B}_1$, and then calculate L, U

Say $L = 11.3$, $U = 14.6$, $\alpha = 0.05$

Then report that " $(11.3, 14.6)$ is a 95% conf. interval for the true regression slope."

Ok: "I am 95% conf. that the true slope is between 11.3 and 14.6."

Not Ok: "There is a 95% prob. that the slope is between 11.3 and 14.6."

Don't write ~~$P(11.3 \leq B_{10} \leq 14.6) = 0.95$~~

Nothing random in that interval.

② The interval is of the form $\hat{B}_1 \pm t_{\alpha/2, n-2} SE(\hat{B}_1)$

and $t_{\alpha/2, n-2} \approx 2$ when $\alpha = 0.05$

Often use $\hat{\theta} \pm 2SE(\hat{\theta})$ as 95% CI

- ★ **Exercise:** Construct a level α hypothesis tests regarding β_1 . Consider both the one-sided and two-sided alternatives.

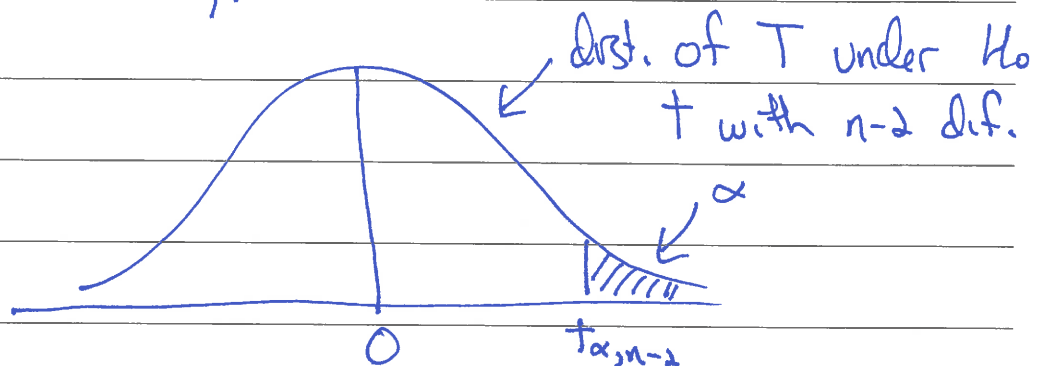
Use $T = \frac{\hat{\beta}_1 - \beta_{10}}{SE(\hat{\beta}_1)}$ as a test statistic.

Under $H_0: \beta_1 = \beta_{10}$, know that T has the t -dist. with $n-2$ d.f.

If testing $H_0: \beta_1 = \beta_{10}$ vs. $H_1: \beta_1 > \beta_{10}$, reject if T is large. In fact, we want to reject if T is larger than $t_{\alpha, n-2}$ since

$$P(T > t_{\alpha, n-2} \mid H_0 \text{ true}) = \alpha$$

Type I error



If testing $H_0: \beta_1 = \beta_{10}$ vs. $H_1: \beta_1 < \beta_{10}$, reject if $T < -t_{\alpha, n-2}$

If testing $H_0: \beta_1 = \beta_{10}$ vs. $H_1: \beta_1 \neq \beta_{10}$, reject if $T > t_{\alpha/2, n-2}$ or $T < -t_{\alpha/2, n-2}$

- 1 A common (overused) objective is to test for evidence if one can claim that
2 $\beta_1 \neq 0$.

3 **Exercise:** Give two ways of using the inference procedures constructed
4 above to address this objective.

5 ① Construct a CI for β_1 . If the
6 interval includes 0, conclude that
7 $\beta_1 = 0$ is among the possibilities at
8 the conf. level

9 ② Perform a hypothesis test of $H_0: \beta_1 = 0$
10 vs. $H_1: \beta_1 \neq 0$
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1 It is important to be mindful of the following fact: If you fail to reject H_0 :
2 $\beta_1 = 0$, or if the confidence interval for β_1 includes zero, this is often not
3 because $\beta_1 = 0$, but instead because the standard error for $\hat{\beta}_1$ is too large to
4 detect the deviation from zero.

5 ★ **Exercise:** How can the standard error of $\hat{\beta}_1$ be decreased?

$$SE(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}_u^2}{(n-1)s_x^2}}$$

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8 Obtain more data to get tests with
9 more "power" to reject H_0
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14 One of the most common mistakes made when constructing models is
15 to blindly remove a predictor from the model solely because there is not
16 strong evidence that its coefficient does not equal zero. Model building (the
17 process of deciding which covariates to include) should comprise many
18 factors, including our knowledge of the situation, the importance that we
19 are placing on constructing a **parsimonious** model, **and** the results of for-
20 mal statistical inference procedures.

The Coefficient of Determination, R^2

When working with regression models it is very common to hear references to “the R^2 for the model.”

This is also called the **coefficient of determination**.

The idea is simple: R^2 equals the proportion of the sum of squared response which is accounted for by the model that has been fit, relative to the model with no covariates.

In particular:

$$R^2 = 1 - \left[\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \right]$$

RSS from model fit
RSS from “null model”

Note that $0 \leq R^2 \leq 1$.

In the case of simple linear regression (a single predictor), R^2 does indeed equal the sample correlation (r) squared.

- 1 R^2 is overused. In particular, note that it is possible for R^2 to be close to
 2 one, but the model be a terrible fit. And it is possible for a great-fitting
 3 model to have R^2 close to zero. See Figure 9.

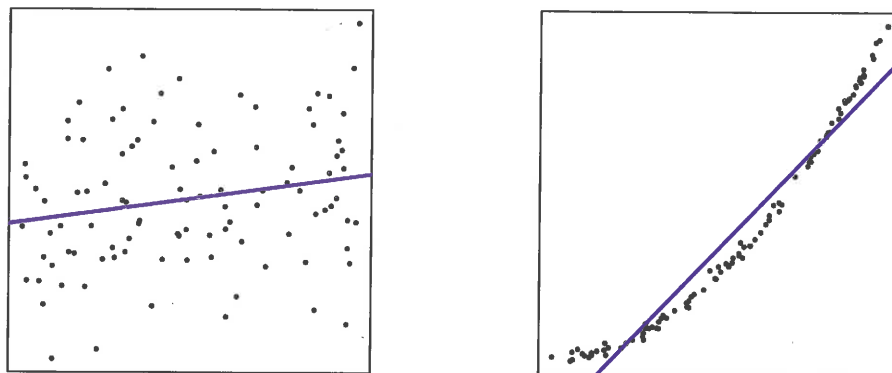
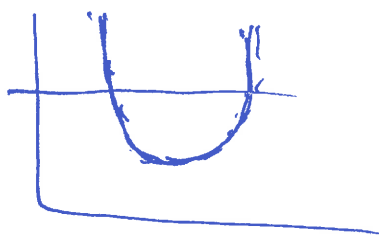


Figure 9: The scatter plot on the left has $R^2 \approx 0.05$, while that on the right has $R^2 \approx 0.95$. The data in the left plot were simulated from the normal linear model.

- 4 In short, R^2 should not be relied upon as a diagnostic to judge the quality
 5 of the model fit. If the model is a good fit, then R^2 says something about
 6 the **predictive power** of the model.



$$R^2 \Leftrightarrow \hat{\beta}_1 = 0$$

Including more predictors will necessarily increase R^2
 So, using R^2 as a model building tool will
 lead to overfitting

* **Exercise:** Suppose that, by mistake, each observed pair (response and predictor) is included twice in the analysis. What is the effect on $\hat{\beta}_i$, the t -statistics, and R^2 ?

$$\hat{\beta}_1 / SE(\hat{\beta}_1)$$

$\hat{\beta}_0$ and $\hat{\beta}_1$ remain the same

$$SE(\hat{\beta}_1) = \hat{\sigma} / S_x \sqrt{n-1}$$

$$\hat{\sigma}^2 = \frac{RSS}{n-2} = \frac{2RSS_{TRUE}}{2n_{true}-2} \approx \left(\frac{RSS_{TRUE}}{n_{true}-2} \right) = \hat{\sigma}_{TRUE}^2$$

S_x^2 stays (almost) the same

but $\sqrt{n-1}$ becomes $\sqrt{2n-1}$, i.e. about $\sqrt{2}$ times larger

So, the $SE(\hat{\beta}_1)$ is incorrectly reported as being divided by a factor of $\sqrt{2}$

$t\text{-stat} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$ is incorrectly multiplied by $\sqrt{2}$

$$S_{xx} = \sum (x_i - \bar{x})^2$$

Diagnostics

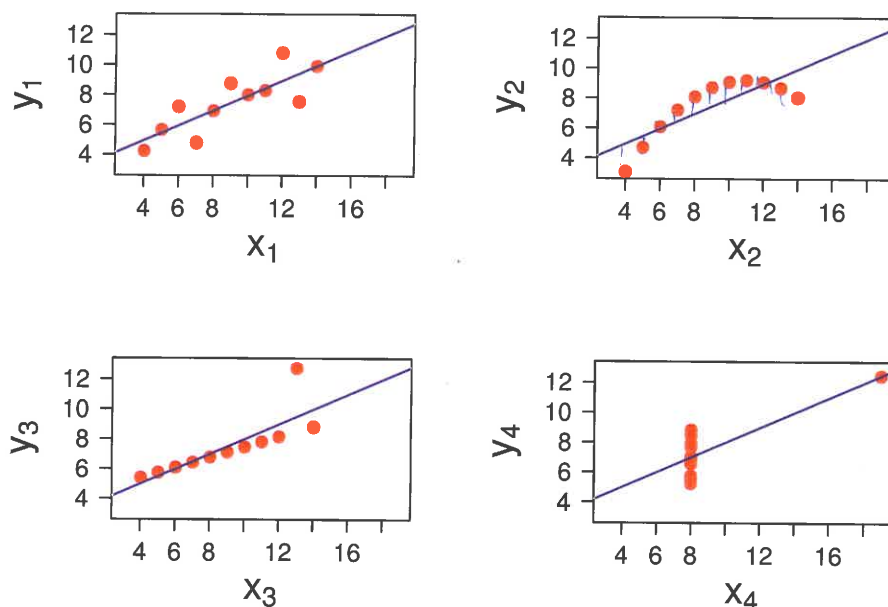


Figure 10: Anscombe's Quartet.

Figure 10 shows four scatter plots constructed by Francis Anscombe for a 1973 research article. These are called **Anscombe's Quartet**. In R, the data for these can be obtained by simply using the command

```
> data(anscombe)
```

Exercise: Use R to find the sample mean and variance for both x and y in all four cases, to find the correlation between x and y in each case, and then use that to find the slope of the regression line for each.

$$\text{cor}(X_1, Y_1) = \text{cor}(X_2, Y_2) = \dots$$

Exercise: This can be taken a step further: We could find that the residual sum of squares in all four cases is exactly the same. What does this imply about the inference for β_1 in all four cases?

All inference procedures give the same results

Exercise: What is the message we take away from this example?

We cannot rely on numerical summaries to assess the quality of model fit

- 1 A very useful diagnostic plot is that of the **residuals versus the fitted values**.
- 2 **ues**. There are a wide range of models for which this is an invaluable tool.
- 3 Figure 11 shows these plots for the four scatter plots of Figure 10.

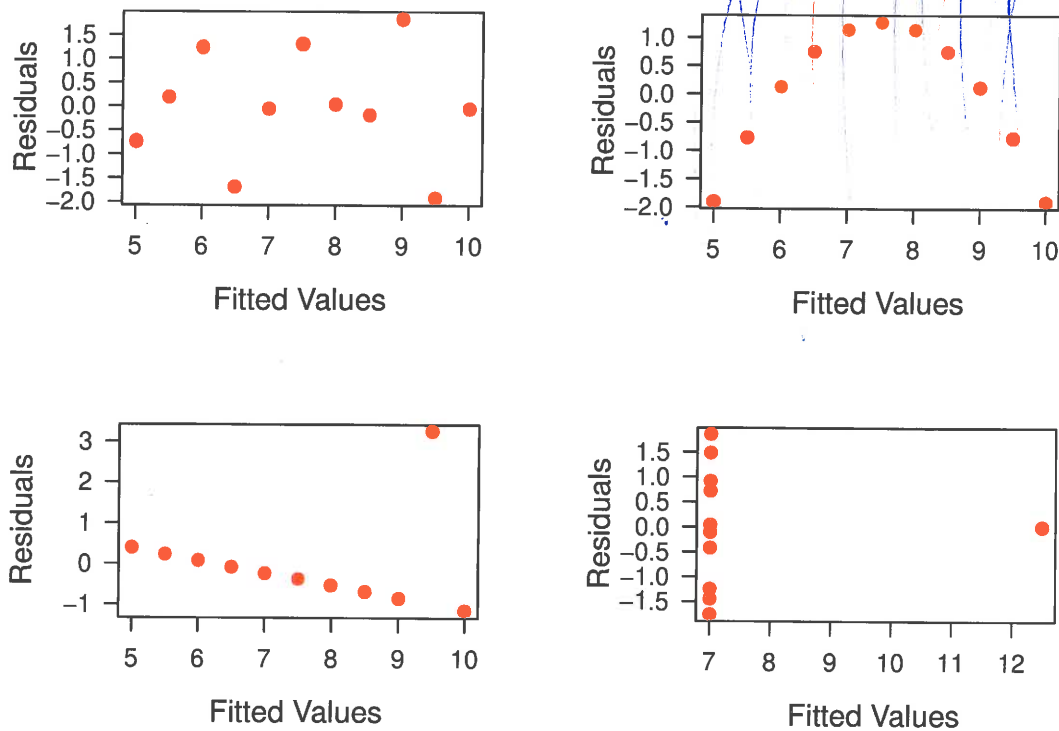


Figure 11: The plots of residuals versus fitted values for Anscombe's Quartet.

- 1 **★ Exercise:** If the assumptions of the model are correct, what should we
2 see in the plot of residuals versus fitted values?

3 There should be no remaining relationship between
4 the residuals and any "fixed" portion
5 of the model fit (predictors, fitted values)
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- 11 **Exercise:** Why is the plot of residuals versus fitted values any more useful
12 than simply looking at the scatter plot?

13 In most cases, have many predictors and hence
14 looking at scatter plot not feasible
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- ★ **Exercise:** Your colleague says, "I found the correlation between the residuals and fitted values to be equal to zero, so there is clearly no relationship between the residuals and fitted values. So, the model must be fine." Do you agree with this logic? No.

$\text{Cor}(\hat{\underline{y}}, \hat{\underline{e}}) = 0$ when β_0 is included in the model

$$\text{cov}(\hat{\underline{y}}, \hat{\underline{e}}) = \frac{1}{n-1} \sum (\hat{y}_i - \bar{\hat{y}})(\hat{e}_i - \bar{\hat{e}}) \quad \leftarrow 0$$

$$= \frac{1}{n-1} \left[\sum \hat{e}_i \hat{y}_i - \sum \hat{e}_i \bar{\hat{y}} \right]$$

$$= \frac{1}{n-1} \left[0 - 0 \right]$$

$$= 0$$